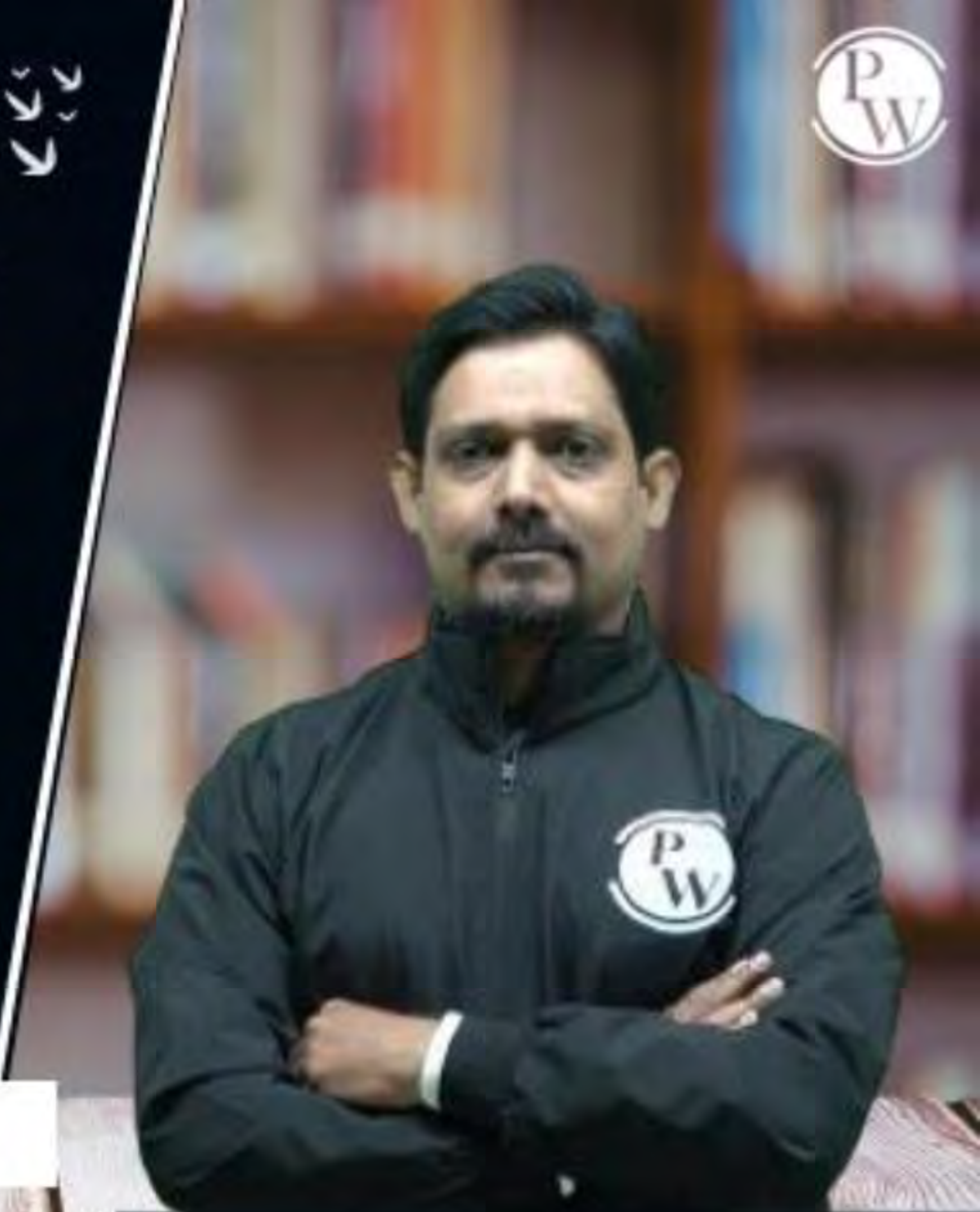
Computer Science & IT

Data Structure & Programming

Tree

Lecture No. 03



By- Abhishek Sir

Recap of Previous Lecture

Topic Binary tree Theorem

Topic Height of Binary tree

Topic Enumeration of unlabeled trees

Topic

Topic

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Topics to be Covered

Topic Counting of Labeled trees

Topic Tree Representation General tree

Topic Binary tree representation

Topic Traversal of Binary tree

Topic

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Number of Unlabeled Trees

$$C_n = \frac{1}{n+1}\,{}^{2n}C_n$$

$C_3 = 5$

← 3 Label with 1,2,3

$3 \times 2 \times 1 = 6$

$3! = 6$

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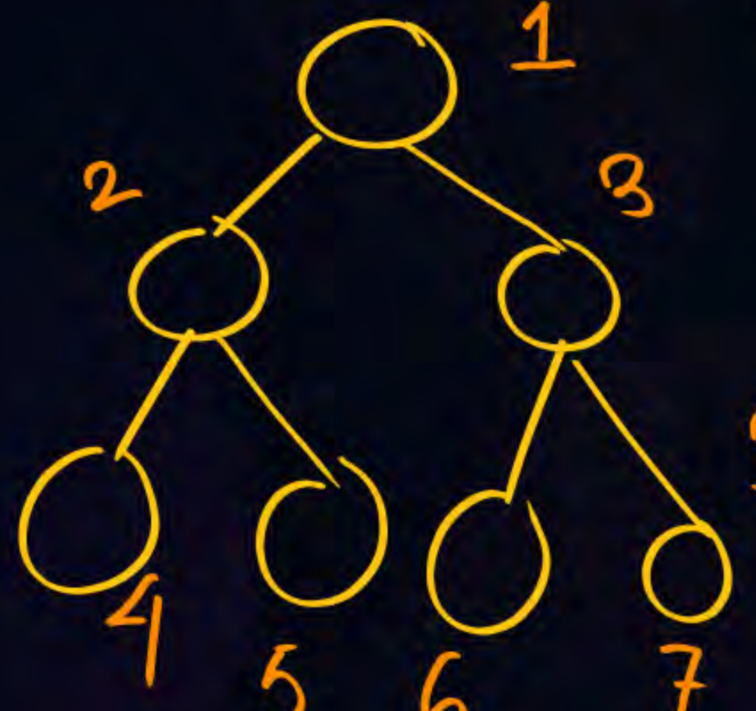
No. of Labeled Binary tree = $\frac{1}{n+1} {}^{2n}C_n \times n!$

Topic : Tree

Sequential Representation: The sequential representation of a binary tree is obtained by storing the record corresponding to node i of the tree as the ith record in an array of nodes.

we use array to store the binary tree



$3\times2=6$

$3\times2+1=7$

Array index starts from 1

if a record is at index i then

Index of Leftchild : $2i$

Index of right child : $2i+1$

Index of parent $(i \neq 1)$ $\lfloor i/2 \rfloor$

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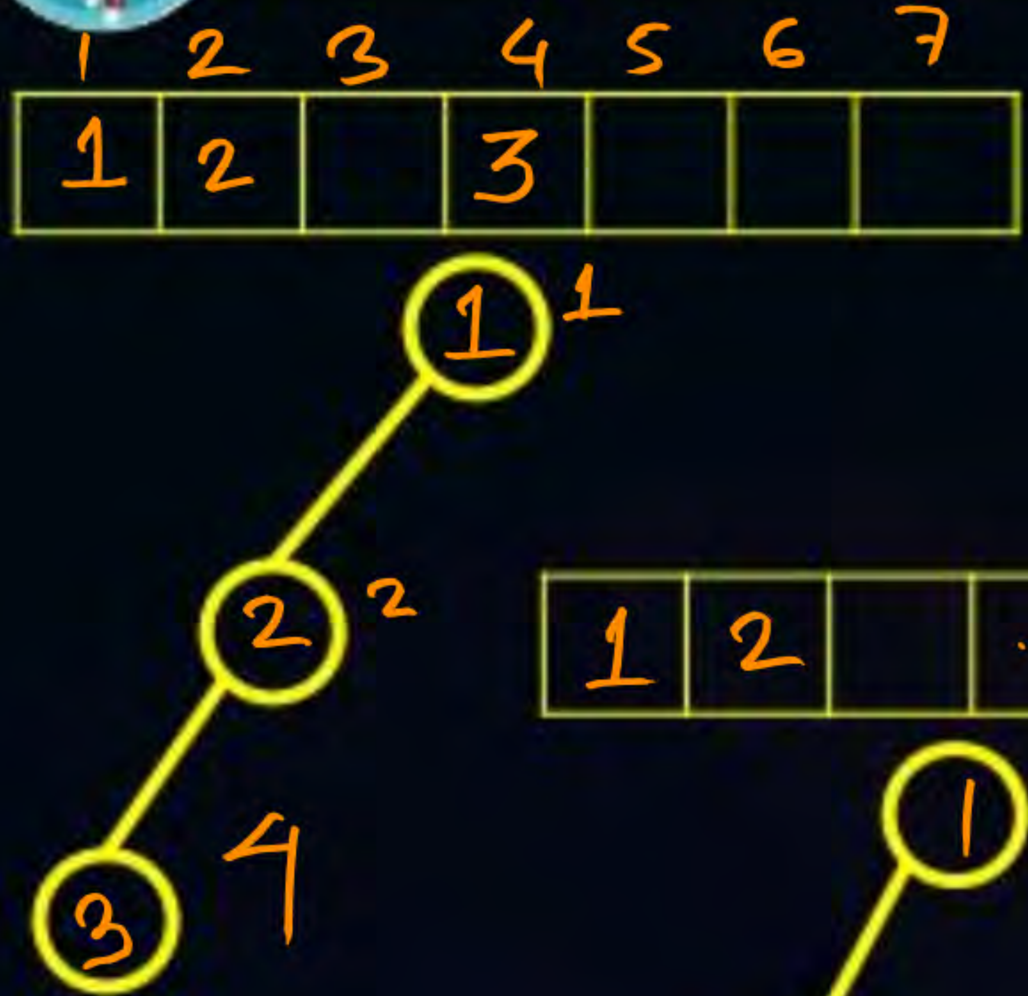
Topic : Tree



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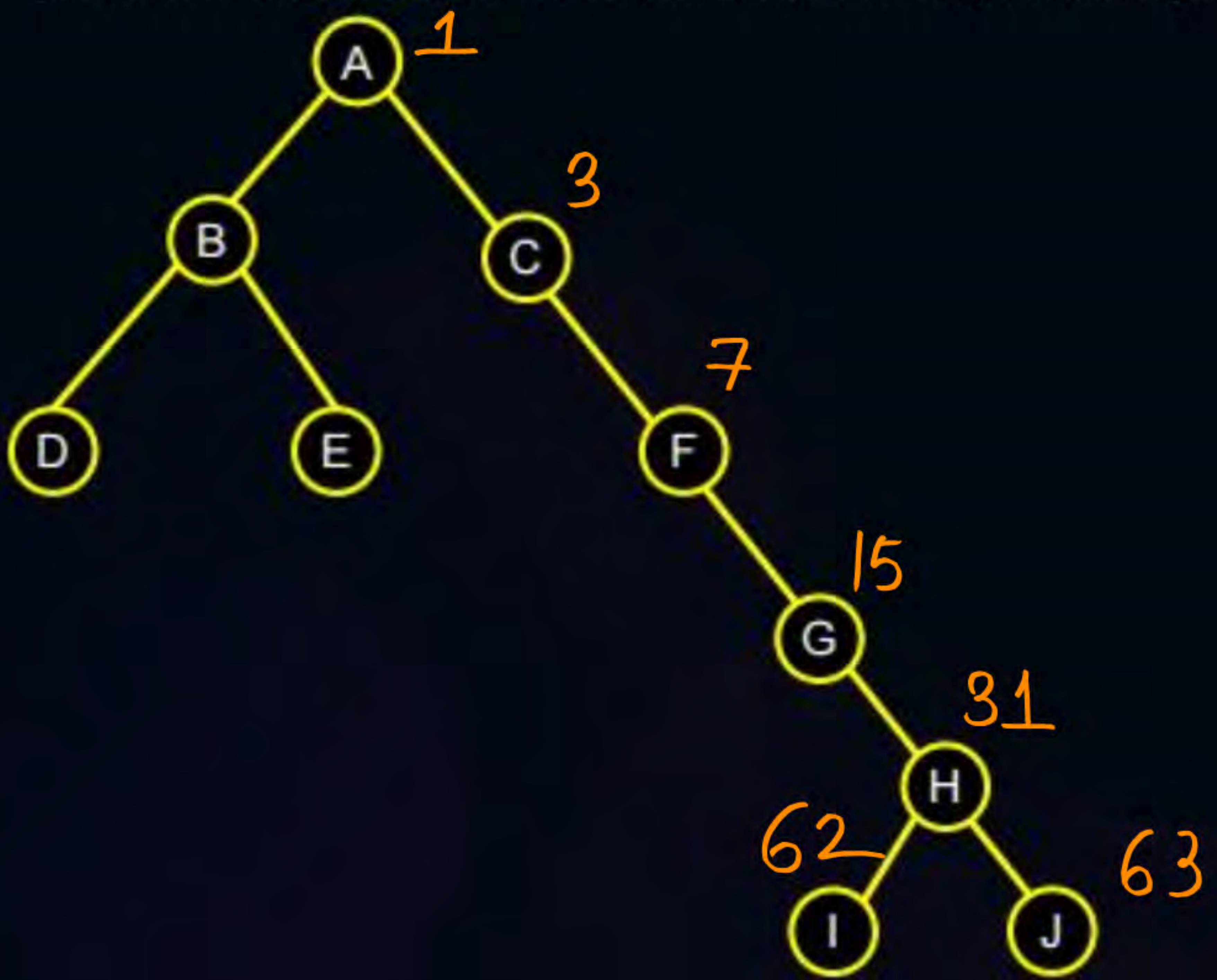
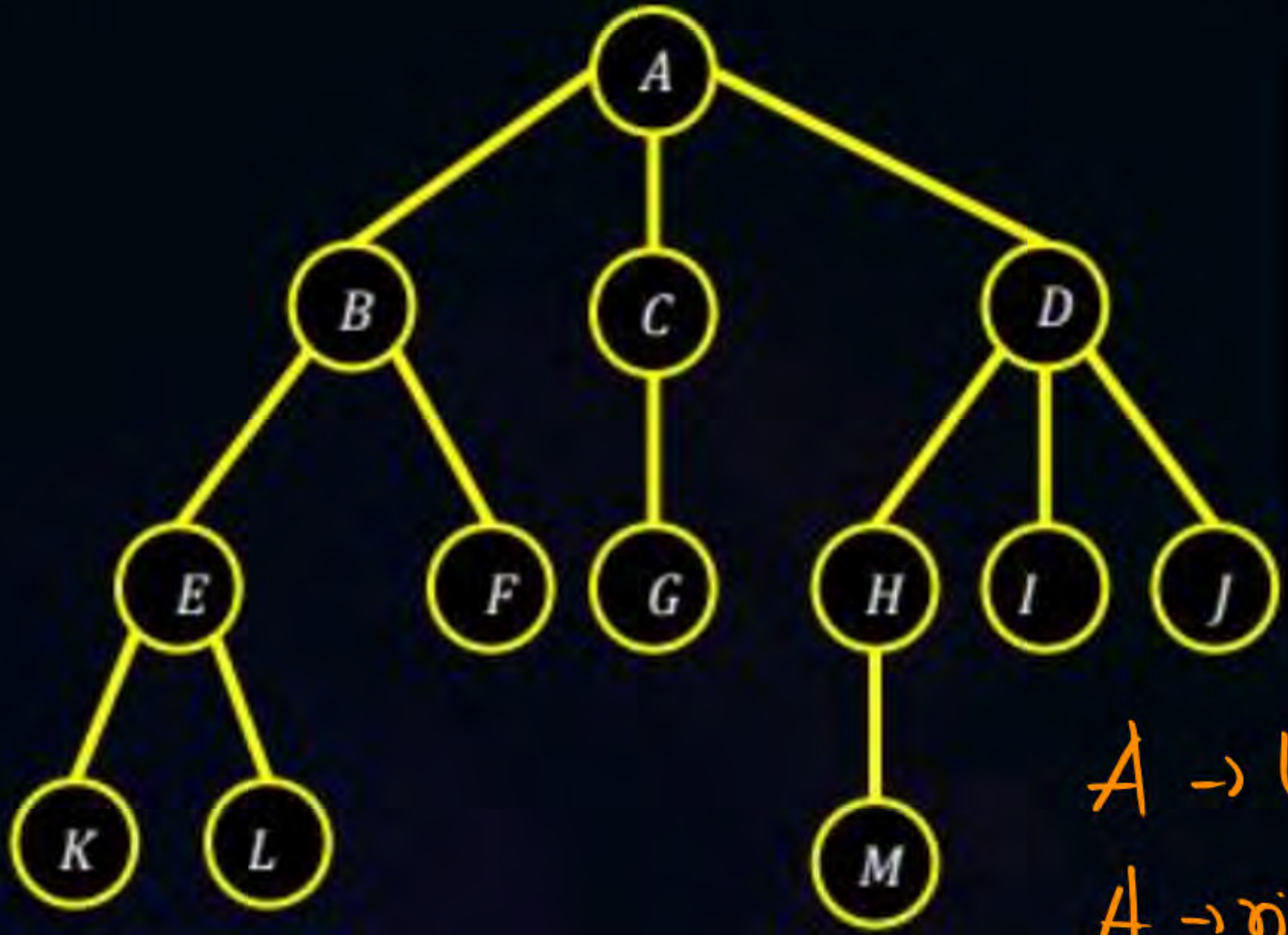
Topic : Tree

#Q If the index start from 1 then index I in sequential representation is The Index of I is.



I

Topic : Generalized Tree Representation



Left most Child - Right Sibling

```
struct Gtnode{
    char ch;
    struct Gtnode * lmchild;
    struct Gtnode * rmsibling;
}
```

A → leftmost child B

A → right sibling - NULL

B - leftmost child - E

B - right sibling : C

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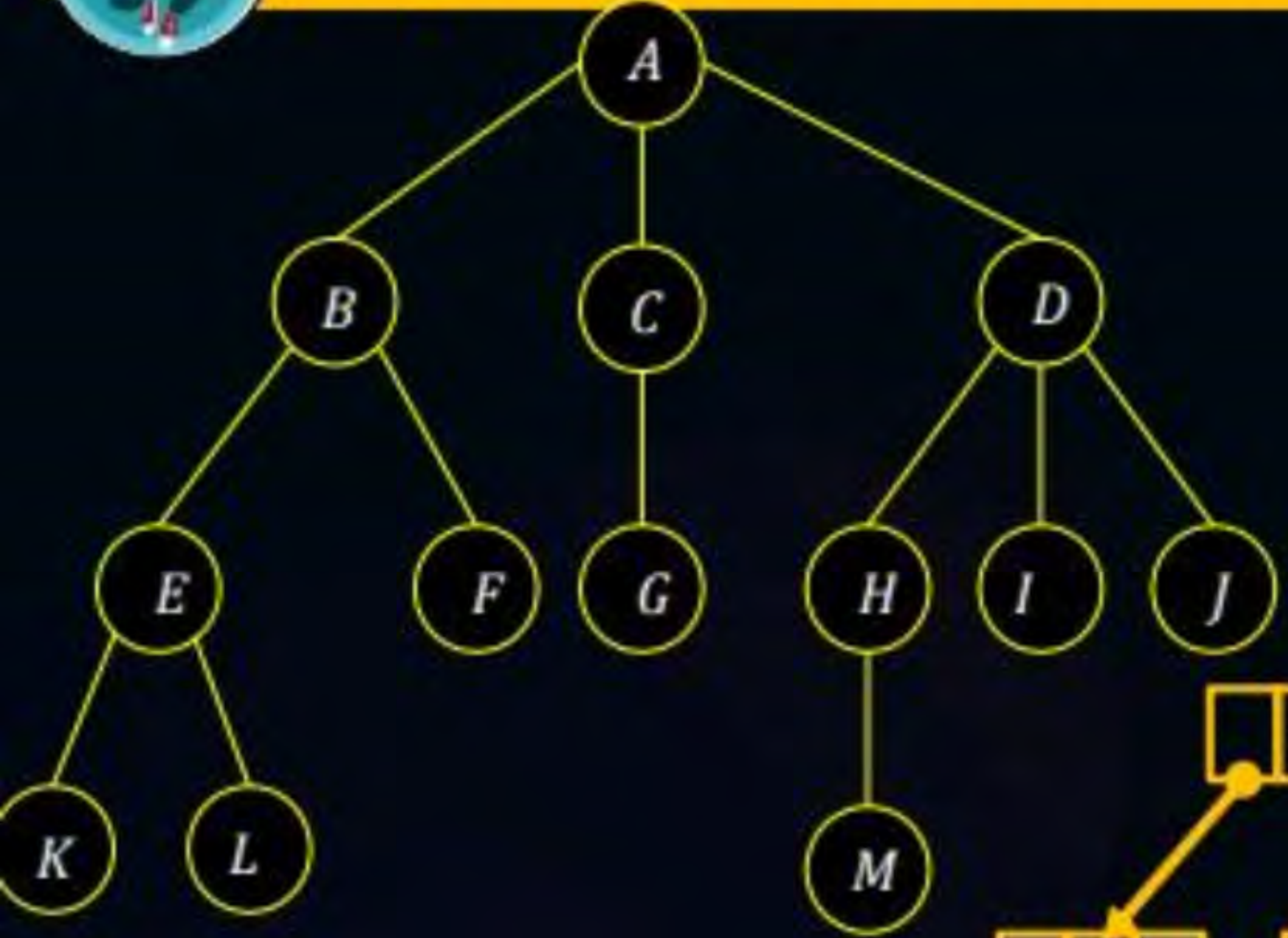
Topic : Generalized Tree Representation



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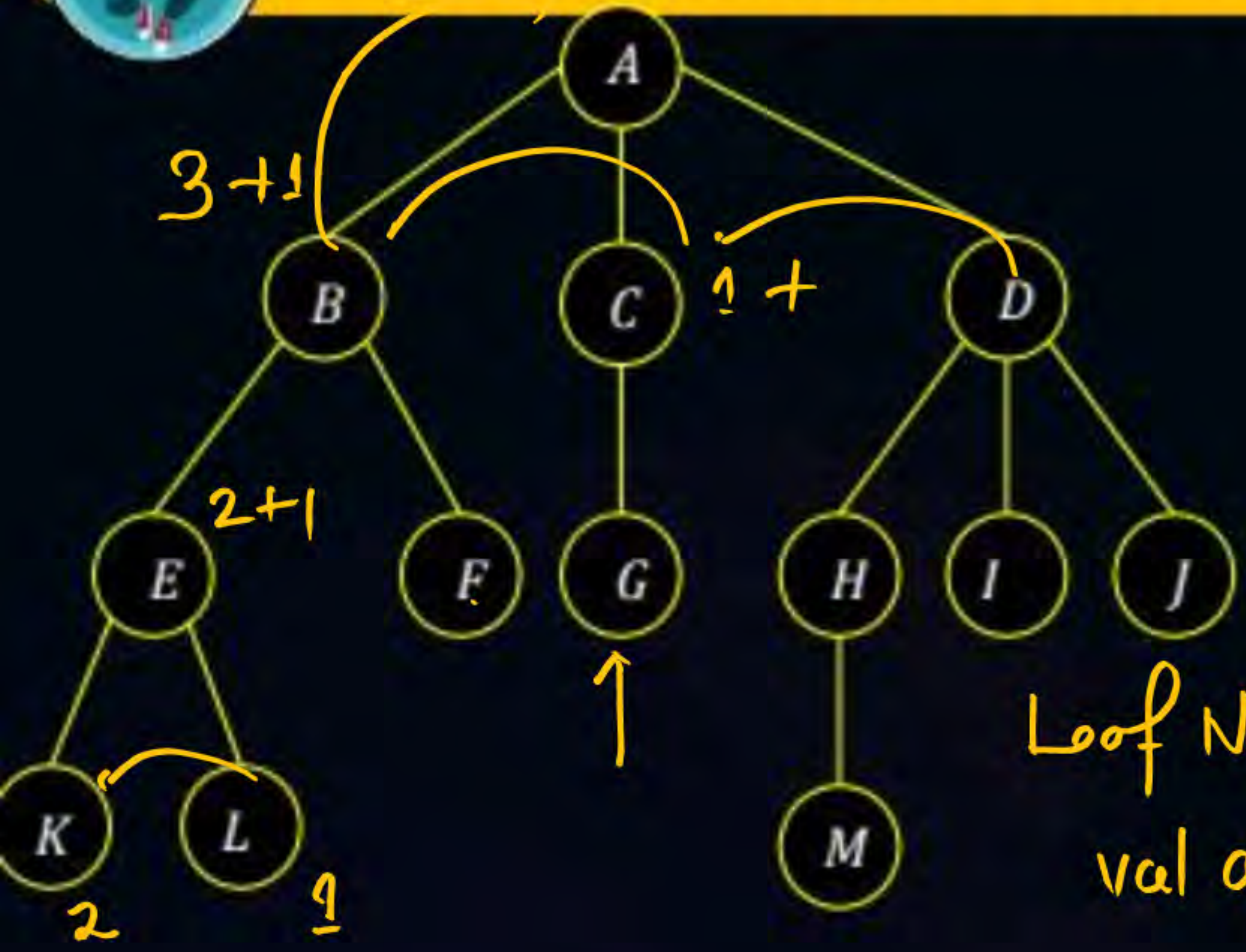
Topic : Generalized Tree Representation



Leaf Node

val assign to 1

```
int foo( struct GTnode * t) {
    int val = 0;
    if(t) {
        if(t→lmchild == NULL)
            val = 1;
        return val + foo(t→lmchild) +
               foo(t→rsibling);
    }
} return 0;
```

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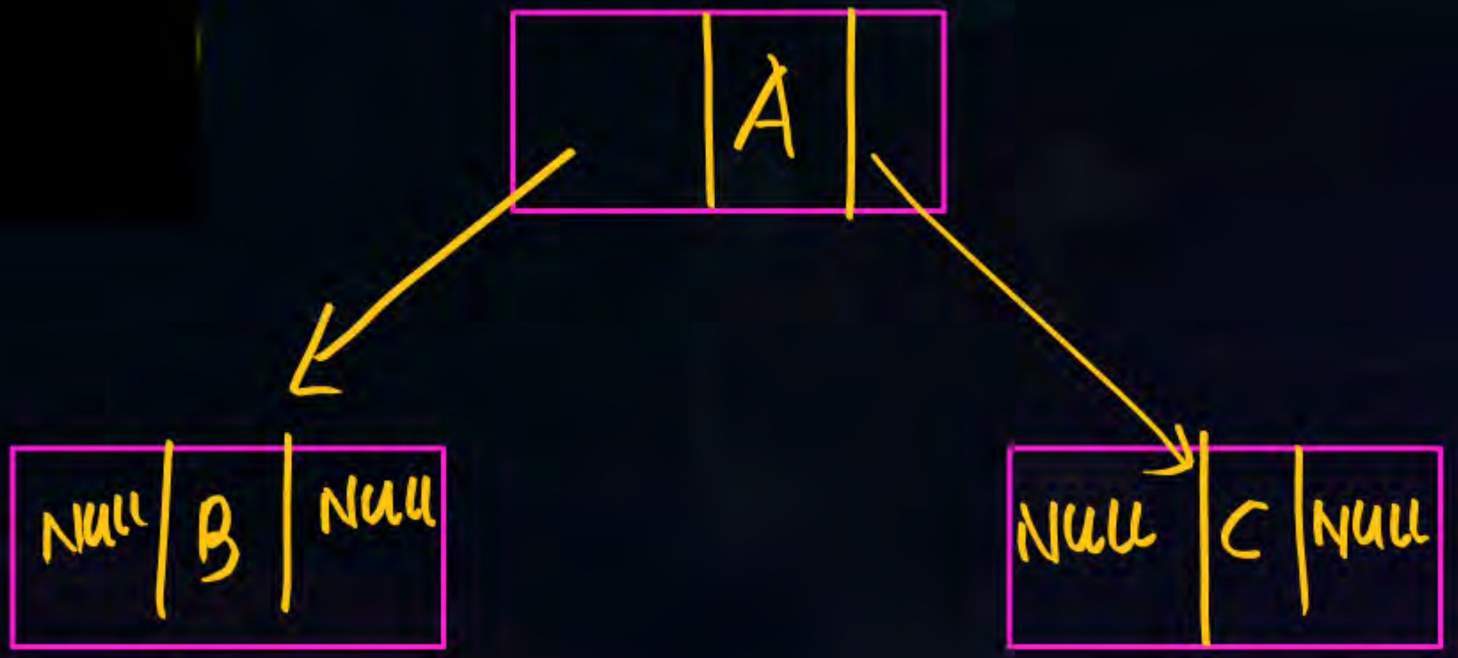
Topic : Tree


```
struct tnode{
    int data;
    struct tnode * lchild;
    struct tnode * rchild;
};
```



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Types of Traversal

Inorder Traversal

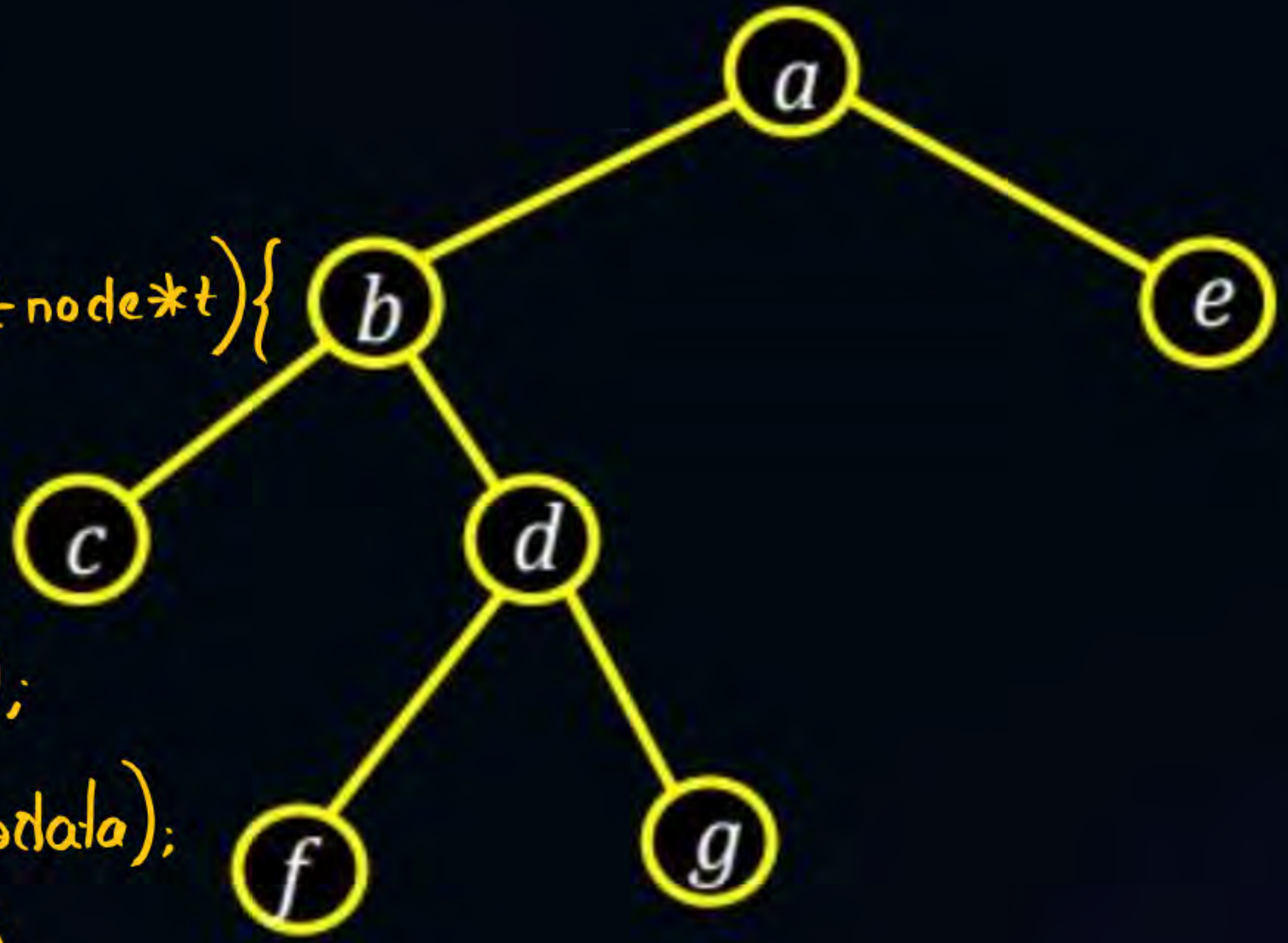
Postorder Traversal

Preorder Traversal

Inorder Traversal

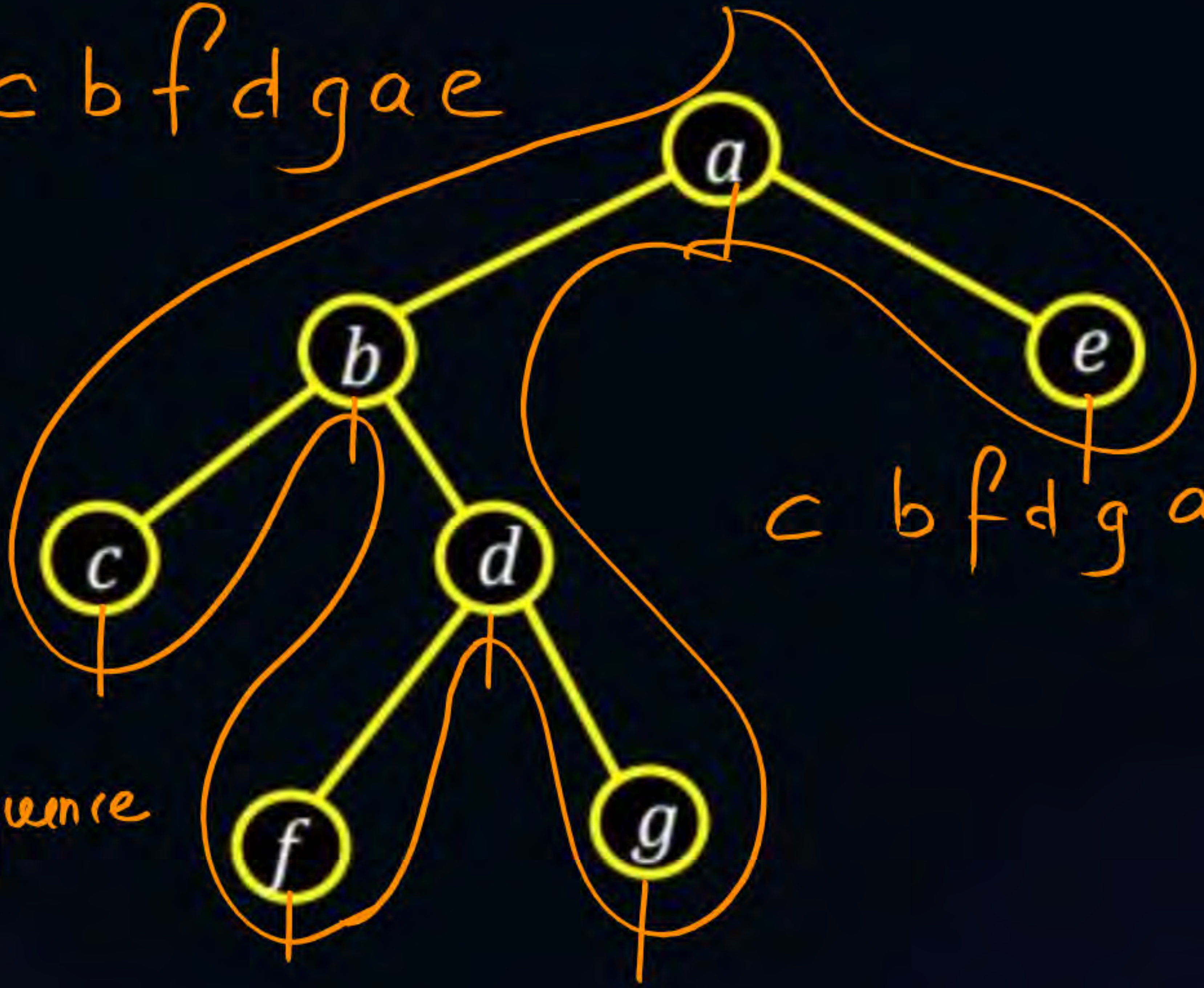
Left Root Right

```
void Inorder ( struct node *t ){
    if (t) {
        Inorder (t→left);
        printf("%d", t→data);
        Inorder (t→right);
    }
}
```



Inorder Traversal

c b f d g a e



c b f d g a e

tracing touches
middle line perform
output the sequence
will be inorder

Preorder Traversal

Algorithm Preorder(tree){

visit the root.

Preorder(left-subtree)

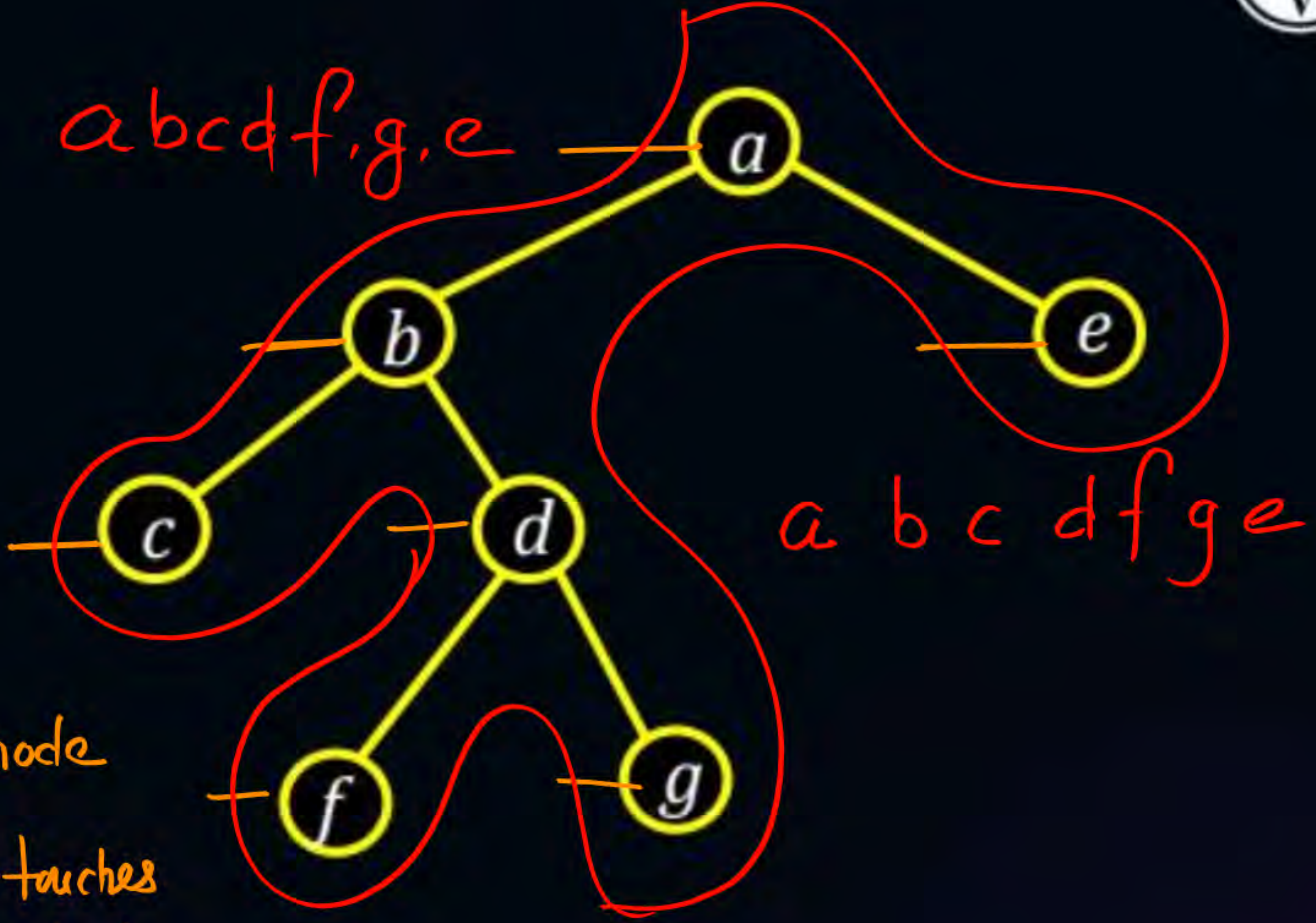
Preorder(right-subtree)

}

Root Left right

draw Left line in node

when the tracing touches right line perform output result in preorder



Postorder Traversal

Algorithm Postorder(tree){

Postorder(left-subtree)

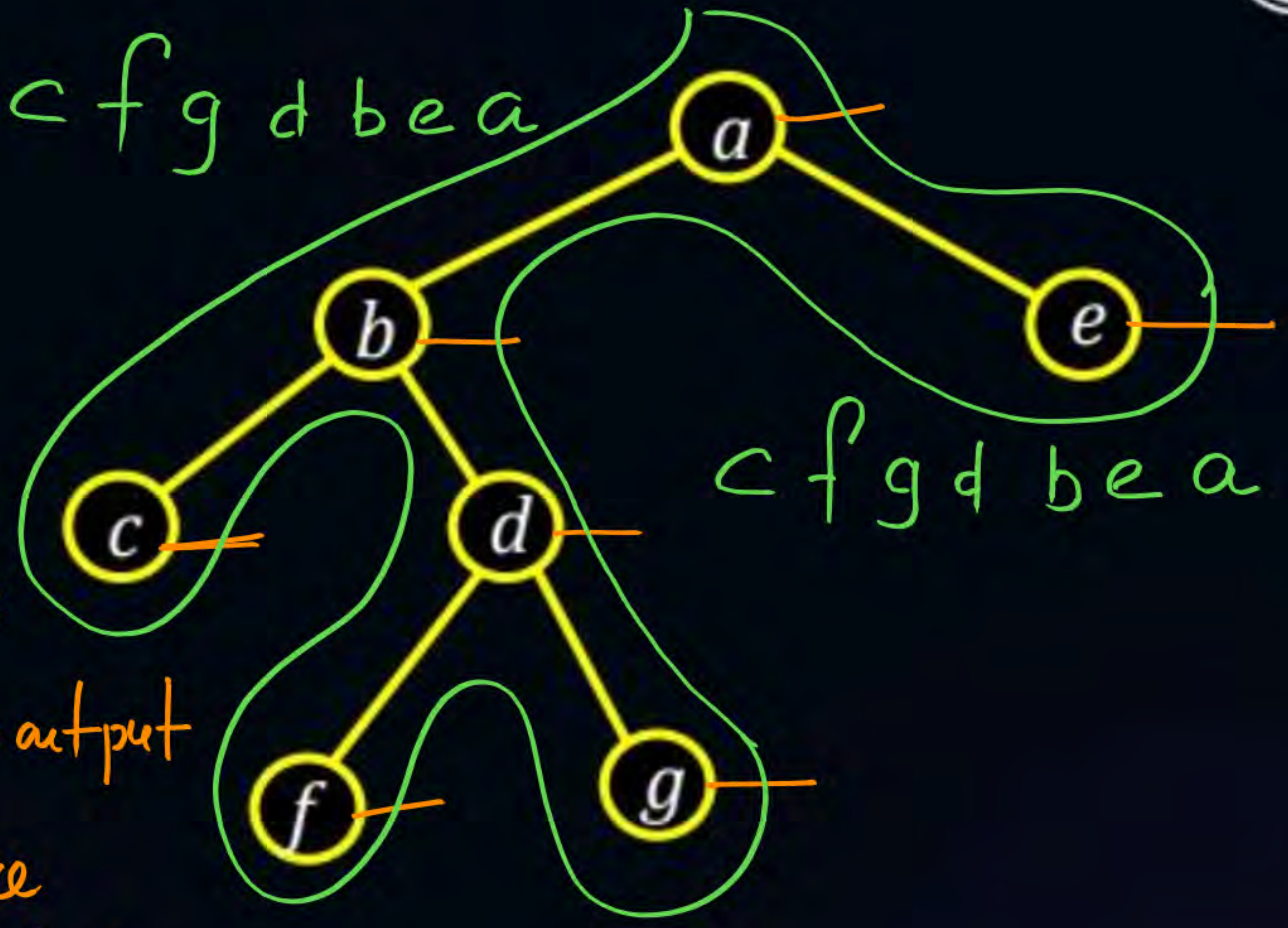
Postorder(right-subtree)

visit the root.

}

Left right root

when tracing touches right line perform output /print the sequence will be postorder



Question

Write Inorder , Preorder and postorder traversal.

In: 2, 7, 6, 11, 1, 9, 5, 8

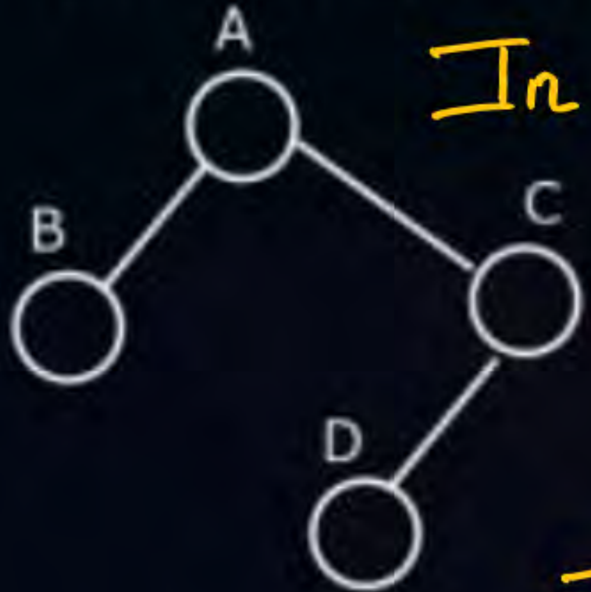
pre: 1, 7, 2, 6, 11, 9, 8, 5

post: 2, 11, 6, 7, 5, 8, 9, 1

Topic : Question

#Q. Which one of the following binary trees has its in-order and preorder traversals as BCAD and ABCD, respectively?

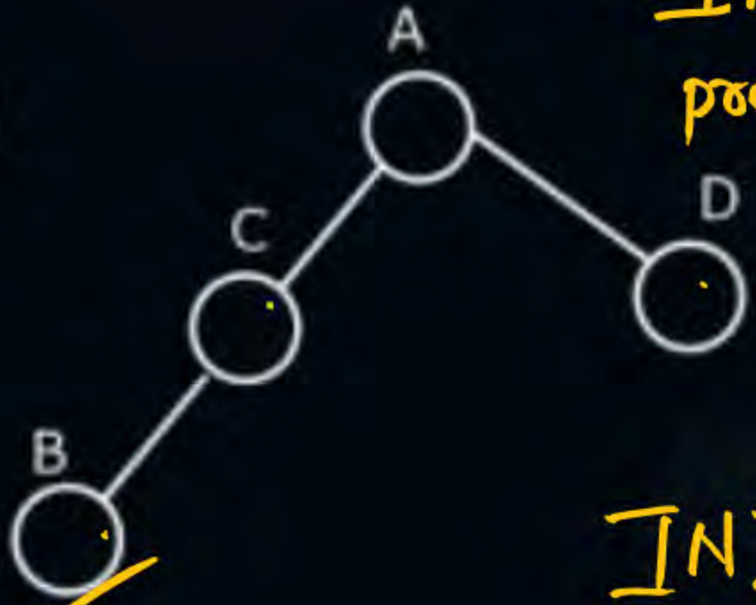
(A)



In BADC

pre: ABCD

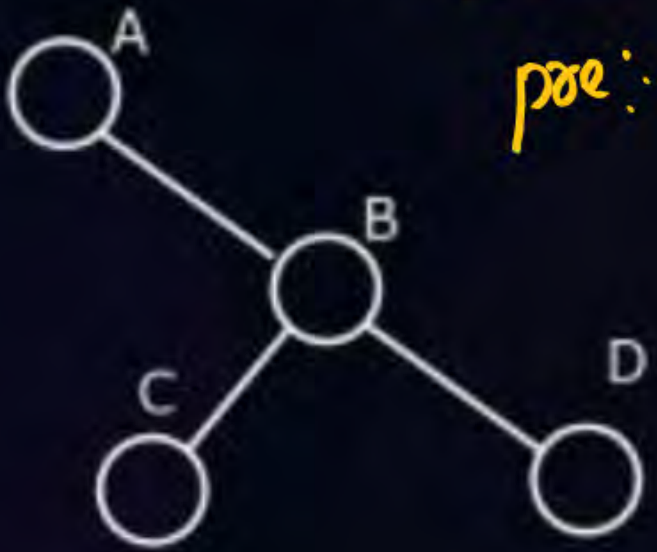
(B)



IN: BCAD

pre: ACBD

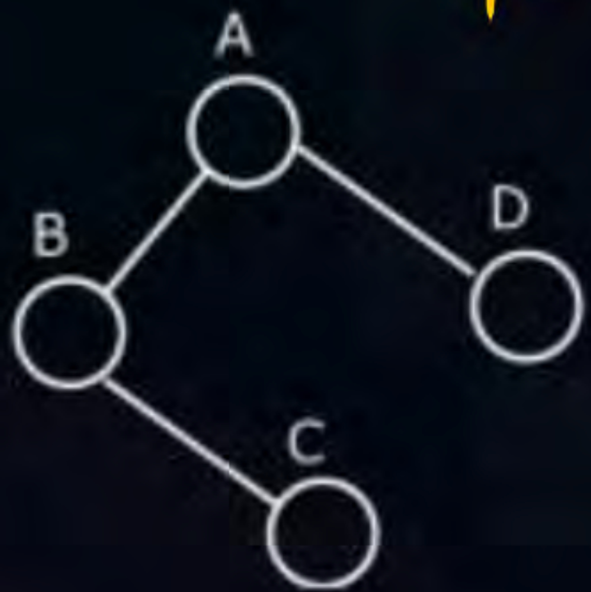
(C)



IN : ACBD

pre: ABCD

(D)



IN: BCAD

pre: ABCD

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Building tree from pre- and in- order

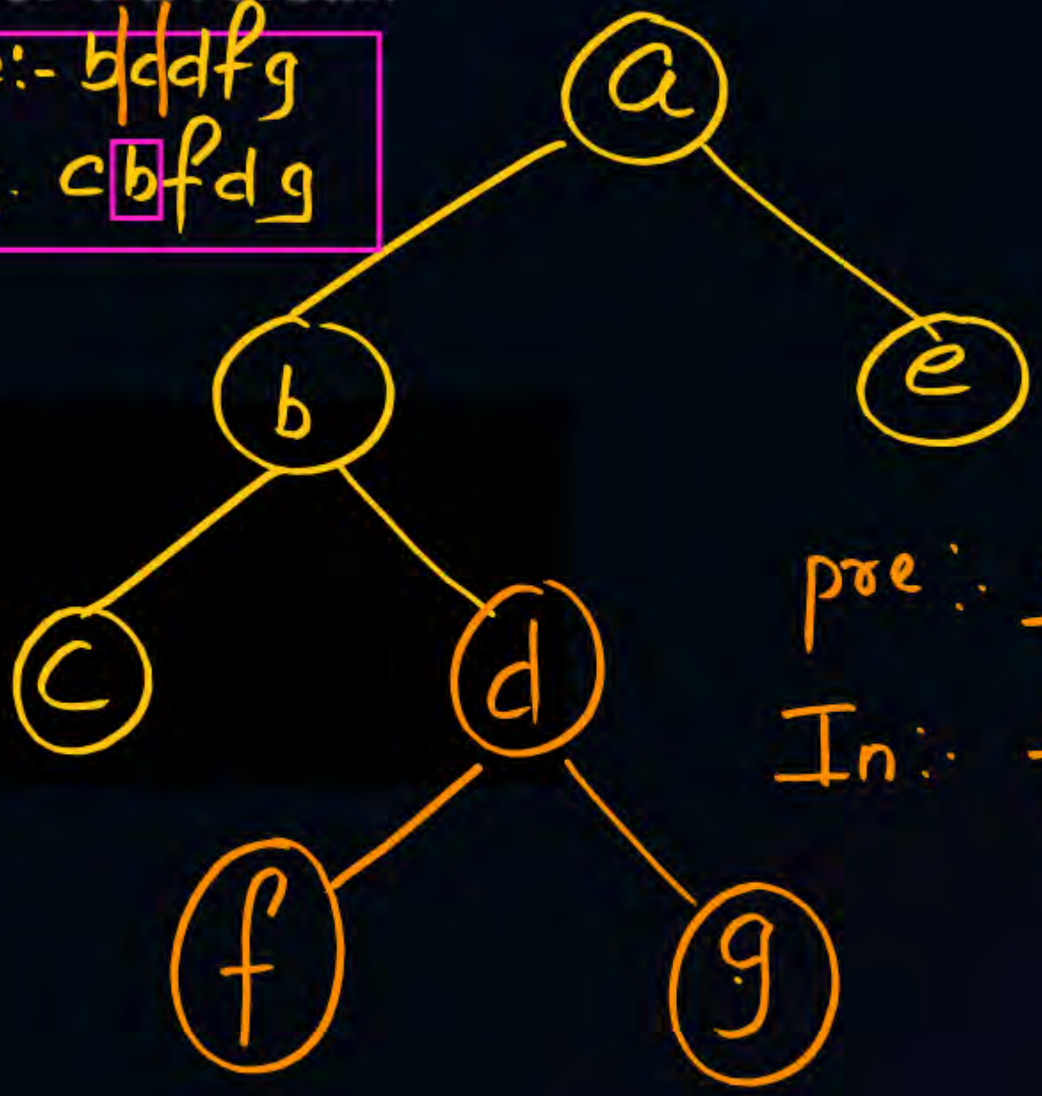
construction of tree with traversal

Construction of binary tree from preorder and inorder traversal.

Preorder: *a | b c d f g | e* Root Left Right

Inorder: *c b f d g | a | e* Left Root Right

pre:- b c d f g
In:- c $\boxed{b}$ f d g



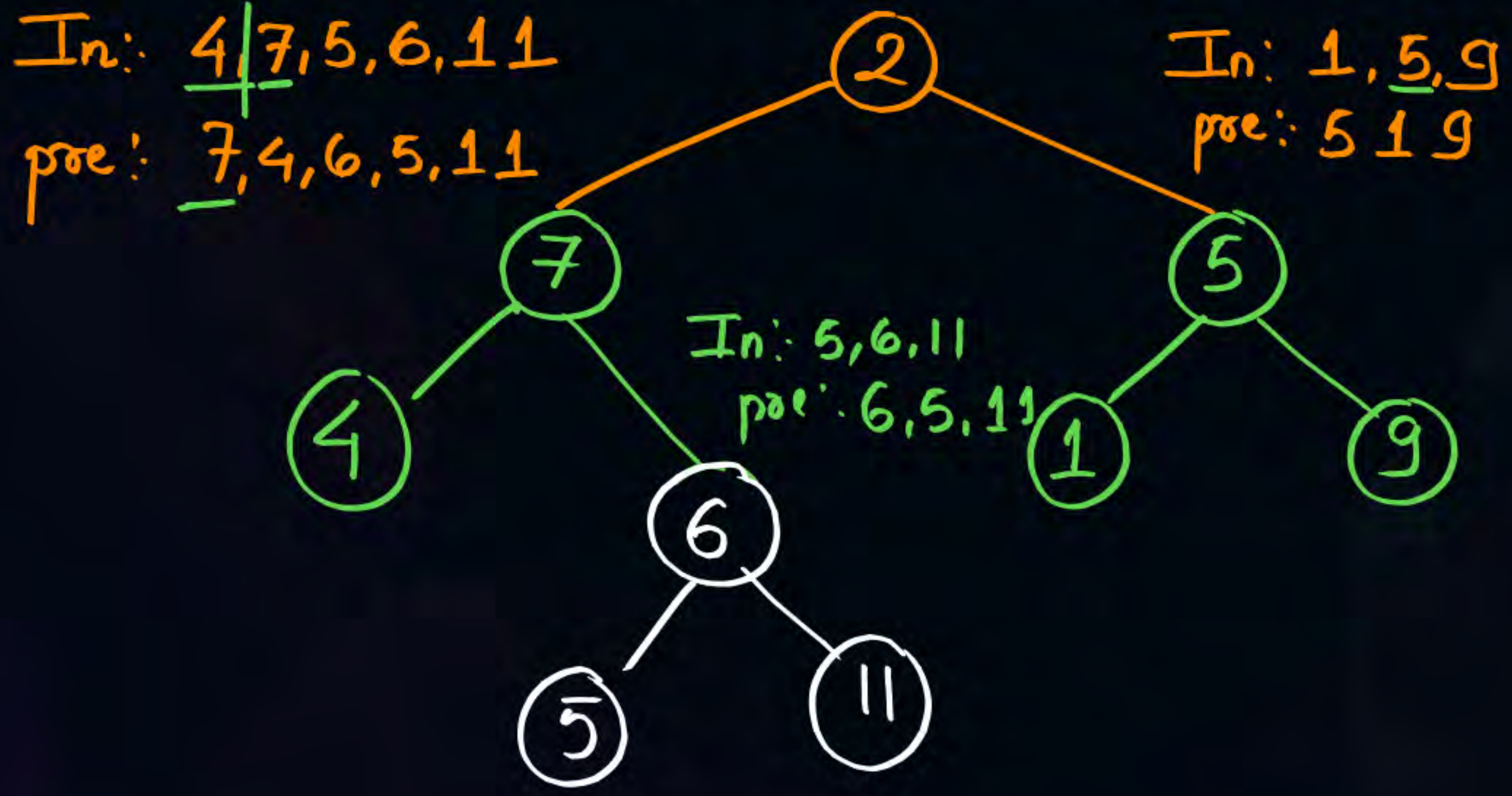
pre:- d f g
In:- f $\boxed{d}$ g

1. Identify the Root from preorder
2. divide the Inorder traversal if left and right subtree based on Root

Topic : Question

#Q if the inorder traversal is 4, 7, 5, 6, 11, 2, 1, 5, 9

pre order : 2, 7, 4, 6, 5, 11, 5, 1, 9. If binary tree constructed then sum of internal nodes is $\boxed{20}$



Ans. 20

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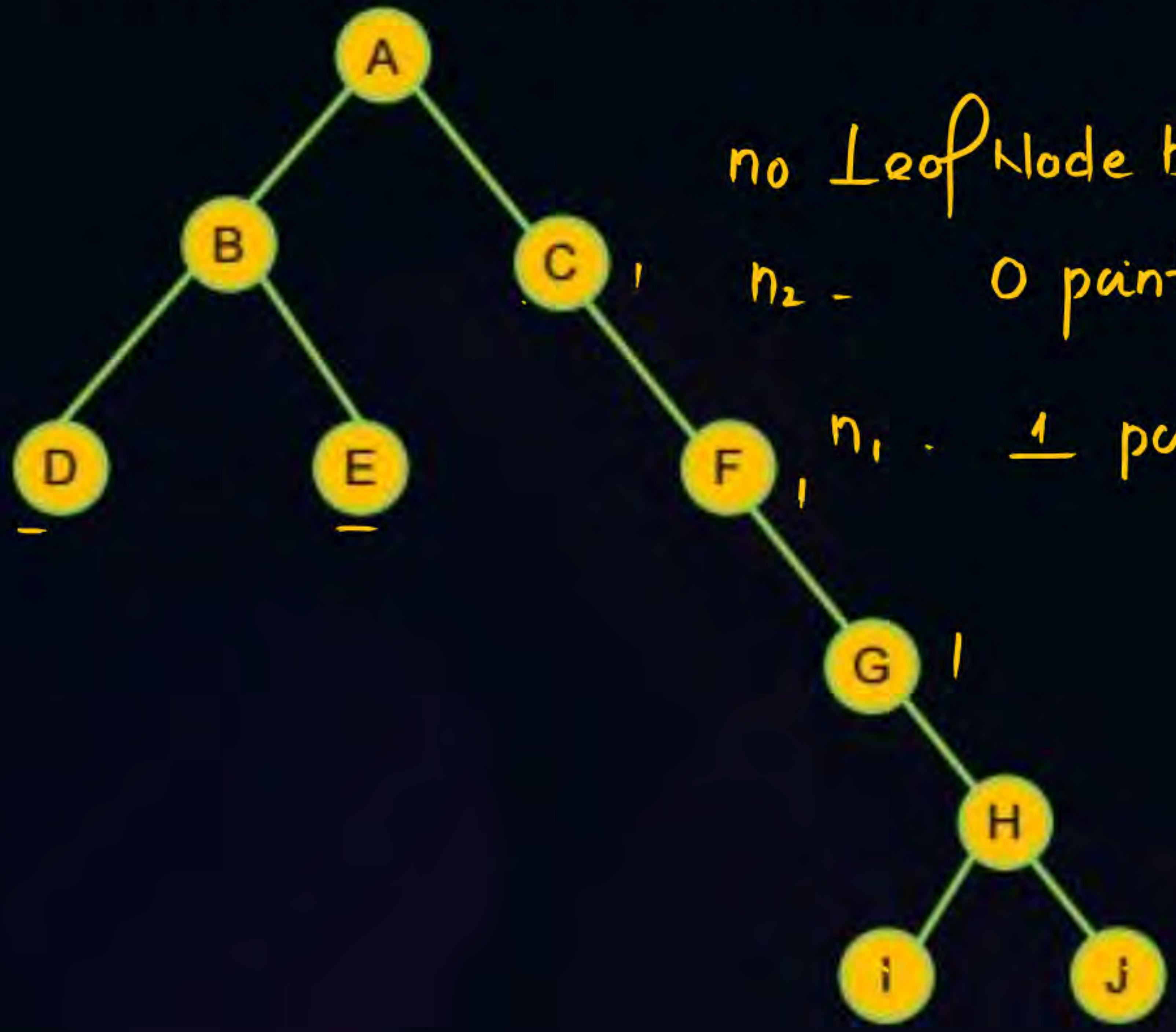
Topic : Tree

\# Number of Null pointer in tree node representation is



tree node representation

no Leaf Node both pointer NULL $4 \times 2 = 8$

n_2 - 0 pointer NULL (2 children)

n_1 - 1 pointer NULL $3 \times 1 = 3$

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Topic : Tree

Q If we use sequential representation of the given tree . The array is having base address of 560 and size is 2B , The address of the element where K will be stored.

$560 + (110 - 1) \times 2$

$560 + 109 \times 2$

$560 + 218$

$= 778$

THAK - YOU

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